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Risk Factors for Injury in Child and Adolescent Sport: A Systematic Review of the Literature

Carolyn A. Emery, BScPT, MSc

Faculty of Kinesiology, Sport Medicine Centre, University of Calgary, Calgary, Alberta, Canada, Department of Public Health Sciences, Faculty of Medicine, University of Alberta, Edmonton, Alberta, Canada

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Objective: The objective of this systematic review of the literature is to identify risk factors and potential prevention strategies that may modify risk factors for injury in child and adolescent sport.

Data Sources: Seven electronic databases were searched to identify potentially relevant articles. A combination of Medical Subject Headings and text words were used (*athletic injuries, sports injury, risk factors, adolescent,* and *child*).

Study Selection: This review is based on epidemiological evidence in which the data are original, an exposure and outcome are objectively measured, and an attempt is made to create a comparison group. Forty-five studies were selected for this review.

Data Extraction: The data summarized include study design, study population, exposures, outcomes, and results. Estimates of odds ratios or relative risks were calculated where study data were adequate to do so. The quality of evidence is based on internal validity, external validity, and causal association.

Data Synthesis: There is some evidence that potentially modifiable risk factors including poor endurance, lack of preseason training, and some psychosocial factors are important risk factors for injury in child and adolescent sport. Concerns with study design, internal validity, and generalizability persist. The evidence is consistent, however, with more convincing evidence from adult population studies. The evidence for nonmodifiable risk factors for injury in adolescent sport (ie, age, sex, previous injury) is consistent among studies.

Conclusions: Sport participation and injury rates in child and adolescent sport are high. This review will assist in targeting the relevant groups and designing future research examining risk factors and prevention strategies in child and adolescent sport. Future clinical trials addressing modifiable risk factors to reduce the incidence of sports injury in this population are necessary.

Key Words: child, adolescent, risk factors, athletic injury

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INTRODUCTION

Sports injuries in children and adolescents may be predictable and potentially preventable.[1,2] Prior to examining potential prevention strategies in child and adolescent sport, we must have a good understanding of the extent of the problem (incidence rates for injury), who is at risk (sport participation), and risk factors for injury in this population.

Participation in physical activity by children and adolescents has important implications for individual and public health benefits. There is epidemiological evidence that better levels of physical fitness are associated with lower all-cause mortality, morbidity, and disease-specific morbidity.[3–5] Child and adolescent sports injury may reduce present and future involvement in physical activity, which may have an impact on future health. Adolescent sport participation rates are high, providing ample opportunity for injury in this population. Based on the 1997 Canadian Population Health Survey, 65% of adolescents reported participation in regular physical activity at least 12 times per month.[6] Similar findings are reported in other countries.[7–11] Canadian adolescents spend on average 14 hours per month participating in physical activity.[12] Although these physical activities may include walking, bicycling, and yard work, a large proportion of Canadian adolescents report participation in sports more prone to injury such as swimming (46%), jogging (44%), basketball (37%), volleyball (26%), and weight training (25%).[12]

Sports injury is the leading cause of injury in adolescents.[8,13–18] Incidence rates reported for studies examining all school-age children (K-12) range from 0.79 to 9.7 injuries/100 children/y.[8,19–24] The reported incidence rate for Canadian grade 10 adolescents is 15.8/100 students/y.[17] Studies that have examined only sport injuries reporting to hospital emergency departments find rates from 7.03 to 8.55 injuries/100 adolescents/y.[13,15,25,26] Sport-specific injury incidence rates reported in studies examining adolescent sports participants range from only 2.6 to 100+ injuries/100

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Reprints: Carolyn A. Emery, BScPT, MSc, Sport Medicine Centre, University of Calgary, 2500 University Drive NW, Calgary, Alberta, T2N 1N4, Canada. E-mail: caemery@ucalgary.ca

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participants/y.[1,5,7,11,27–46] In adolescent studies that have examined level of exposure, sport-specific incidence densities reported range from 2.38 to 142.86 injuries/1000 participation hours.[7,23,28,36,39,47–55] Sport-specific rates of injury vary considerably, with the highest rates of injury reported for boys participating in hockey,[28,41] basketball,[7,24,56] and football[24,31,36] and for girls participating in gymnastics,[13,24,36] basketball,[7,56] and soccer.[7,49,54,56] The lowest rates of injury are consistently reported in swimming, tennis, and badminton.[7,24,27,56]

Acute trauma is 1 type of injury sustained in child and adolescent sport. In addition, there is growing concern about overuse injury in this population of athletes.[57] This likely reflects increased intensity of training and competition in sport at younger ages, increased skill level at younger ages, and longer (often year-round) training seasons.[57]

Risk factors in sport are any factors that may increase the potential for injury.[2] Risk factors may be extrinsic (ie, weather, field conditions) or intrinsic (ie, age, conditioning) to the individual participating in the sport. Modifiable risk factors refer to those that have the potential to be altered by injury prevention strategies to reduce injury rates.[2] Nonmodifiable risk factors, which cannot be altered, may affect the relationship between modifiable risk factors and injury. Identification of these factors will assist in defining high-risk populations. Potential risk factors (adapted from Lysens et al[1] and Caine et al[58]) are listed in Table 1.

Much of the literature addressing child and adolescent sports injury is sport-specific and based on descriptive data that portray primarily the extent of the injury problem. Reviews in the literature examining injury risk often have not been systematic in nature (ie, lack presentation of search strategies or assessment of study validity), typically address individual sports, and include adult studies. Arguably, there is a need for a global, comprehensive review of risk factors for injury in child and adolescent sport to provide direction for further research in injury prevention in this population.

TABLE 1. *Potential Risk Factors for Injury in Child and Adolescent Sport*

Extrinsic Risk Factors	Intrinsic Risk Factors
Nonmodifiable	Nonmodifiable
Sport played (contact/no contact)	Previous injury
Level of play (recreational/elite)	Age
Position played	Sex
Weather	
Time of season/time of day	Potentially Modifiable
	Fitness level
Potentially modifiable	Preparticipation sport-specific training
Rules	Flexibility
Playing time	Strength
Playing surface (type/condition)	Joint stability
Equipment (protective/footwear)	Biomechanics
	Balance/proprioception
	Psychological/social factors

The objective of this systematic review is to identify risk factors for injury in child and adolescent sport as well as potential prevention strategies that may modify risk factors and reduce injury in this population. Studies that have examined an association between risk factor and injury or a prevention strategy and injury in child and adolescent sport are reviewed and assessed regarding their validity, generalizability, and strength of scientific evidence (ie, based on study design, validity, and causality). Recommendations are made for future epidemiologic research to assist in determining strategies for prevention of injury in child and adolescent sport.

METHODS

Data Sources

Seven electronic databases were searched by the author to identify potentially relevant articles. These databases included Medline (1966–2001), CINAHL (1982–March 2002), Psychinfo (1967–present), Cochrane database for Systematic and Complete Reviews, the Cochrane Controlled Trials Registry, HealthSTAR (1975–present), and sportdiscus (1980–2001). A combination of Medical Subject Headings (MeSHs) and text words were used in this search. MeSHs and text words included *athletic injuries*, *risk factors*, *adolescent*, and *child*. An additional text word that was not a MeSH was *sports injury*. No limitations were put on articles searched. If 500 articles or fewer were identified by a given search strategy, the study title and abstract were reviewed to identify articles potentially relevant to the subject area. The methods sections of potentially relevant articles were then reviewed to identify studies that met the selection criteria.

Study Selection

Inclusion criteria were the following:

(1) Report is a human epidemiological study that assesses the association between any potential injury risk factor or prevention strategy and injury in child and adolescent sport
(2) Outcome includes a measure of injury sustained in sport
(3) Exposure measure includes some objective measurement of risk factor or intervention
(4) Study design includes a comparison group (cross-sectional, case control, cohort, quasi-experimental, and randomized clinical trial [RCT] designs all considered)
(5) Study contains original data

Exclusion criteria were the following:

(1) Sport-related injury involves the following high-speed sports: bicycling, scootering, skateboarding, inline skating, tobogganing, skiing, snow boarding, and boating
(2) Study reports exclusive examination of head or dental injuries or medical emergencies

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(3) Prevention strategy involves protective equipment (ie, helmets, knee braces) to modify risk of sport injury

The rationale for exclusion criteria were that risk factors and prevention strategies involving particular high-speed sports and protective equipment were specialized enough to be addressed independently and were beyond the scope of this review.

Data Extraction

The data extracted included study design, study population, exposure of interest measured, outcome measure, and results. Point estimates (including 95% confidence intervals [CIs]) of odds ratios (ORs) or relative risks (RRs) were calculated where study data was adequate to do so if these were not reported in the reviewed studies. The quality of evidence was assessed based on criteria regarding internal validity, external validity, and Hill criteria for causal association (Table 2).[59]

The internal validity of each study was assessed based on the following criteria:

(1) Strength of study design
(2) Presence of selection bias
(3) Presence of misclassification bias, including an appraisal of reliability and validity of measurements (exposure and outcome)
(4) Control of potential confounding

Data Synthesis

Characteristics of the studies are summarized, including study design, study subjects, exposure measured, outcome measured, and results. Assessment of the studies with respect to internal validity, external validity, and causal association is summarized. Findings of each study in addition to relative strengths and weaknesses are compared. Comparisons with adult population studies are reviewed. Specific recommendations are made for future research.

RESULTS

The findings of the Medline search are described in Table 3. Twenty-five of the potentially relevant studies identified were included in the review. The findings of the sportdiscus search identified an additional 45 potentially relevant articles, of which 5 were included. Fifty-four further articles were identified in the CINAHL search, of which 7 were included. No further studies were identified in the additional 4 databases. In total, 37 studies were chosen to be included in this systematic review from the search strategy described. In addition, references from 2 key sport medicine texts including epidemiological reviews were examined.[58,60] An additional 9 relevant articles were included in this systematic review, bringing the total to 46. A summary of research design, subjects, exposure measures, outcome measures, and results for all 46 studies reviewed is found in Table 4.

TABLE 2. *Hill Criteria for Causality as Defined in Rothman*[59]

Criterion	Definition
(1) Strength	Strong associations more likely to be causal
(2) Consistency	Repeated observation of association in different populations under different circumstances
(3) Specificity	Cause leads to a single effect, not multiple effects
(4) Temporality	Cause precedes an effect in time
(5) Biologic gradient	Presence of a unidirectional dose-response curve
(6) Plausibility	Biologic plausibility of the hypothesis
(7) Coherence	No conflict with what is known of the natural history and biology of the disease
(8) Experimental evidence	Basic science and/or epidemiologic human experiments
(9) Analogy	Draws a parallel with an understood association (provides a source of more elaborate hypotheses about the association under study)

Risk Factors for Injury in Child and Adolescent Sport

Nonmodifiable Risk Factors

In identifying nonmodifiable risk factors for injury in child and adolescent sport, there is evidence that males are generally at greater risk for injury (OR = 1.16–2.4.)[6,8,20,21,24,30] The exception to this is in studies examining specific sports, including soccer,[20,45,47,53,54] baseball,[61] and basketball,[61] in which females appear to be at greater risk. Left-handedness also appears to be a risk factor for injury.[62] Reinjury rates range from 13.1% to 38%.[1,7,31] The risk of reinjury in football is greater than the risk of first-time injury (RR = 1.4–1.7).[42,63] Sport-specific rates of injury vary considerably, with the highest rates of injury reported for boys participating in hockey,[28,41] basketball,[7] and football[31,36] and for girls participating in gymnastics,[13,36] basketball,[7] and soccer.[49,54,55]

Risk of injury consistently increases with age across studies.[7,8,24,28,31,39,42,43,45,47,52,55,64,65,66] Consistently, in all sports, adolescents (>13 years) are at a greater risk of injury than younger children.[7,8,24,28,31,47,52,55] The peak injury rate is consistently in the oldest adolescent age group studied in reports examining all sports, soccer, hockey, football, baseball, and gymnastics.[7,8,24,28,31,47,52,55] Injury rates decrease with increasing skill level in hockey[28] and increase with increasing skill level in wrestling.[39] Risk of injury increases with organized sport versus unorganized sport,[24] amount of time spent doing sporting activity,[30] competition versus practice,[39,40] tournament play versus regular season play,[41,54] increased level of competition,[7] indoor versus outdoor soccer,[64] and large field size and reduced number of players in Australian rules football.[52]

There is conflicting evidence regarding anthropometric measurement and risk of injury, which appears to be injury and sport-specific. Brust et al[28] demonstrate an increased risk of injury in lighter hockey players with the same age and experience. In football, however, where age categories are also restricted by weight categorization, heavier players are at higher risk of injury than boys who are lighter.[31,44,52,67] In gymnastics, athletes who are taller or heavier are at an increased risk of injury compared with those shorter or lighter.[43,66] In soccer, Back-

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TABLE 3. *Medline Strategy and Findings*

MeSH or tw	MeSH or tw Defined	No. Articles Identified*	No. Articles Determined as Potentially Relevant†	No. Articles Included in Critical Appraisal‡
(1) MeSH or tw	Athletic injuries or "athletic injuries"	11,889	n/a	n/a
(2) MeSH or tw	Athletic injuries or "sports injury"	11,918	n/a	n/a
(3) MeSH or tw	Risk factors or "risk factors"	227,841	n/a	n/a
(4) MeSH or tw	Adolescence or "adolescent"	946,841	n/a	n/a
(5) MeSH or tw	Child or "children"	1,742,010	n/a	n/a
1 and 3 and 4		386	52	19
2 and 3 and 4		387	52	1 additional
2 and 3 and 5		408	48	5 additional

* Number of articles identified by given strategy in Medline search.

† Number of articles identified by author as potentially relevant to subject area based on abstract

‡ Number of articles identified, following review of methods section in article, as relevant for critical appraisal based on inclusion criteria

MeSH indicates Medical Subject Heading; n/a, not assessed >500; tw, textword.

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ous et al[47] demonstrate that taller players are at an increased risk of injury compared with shorter players. Lyman et al[65] demonstrate increased risk of elbow symptoms in pitchers who are heavier and taller.

Potentially Modifiable Risk Factors

Twenty of the 46 epidemiological studies reviewed examine potentially modifiable physical intrinsic risk factors specifically for injury in child and adolescent sport.[1,29,33,36,39,43,47,52,64–66,68–76] Most studies examining biomechanical alignment, flexibility, or strength demonstrate no association of these factors with injury in child and adolescent sport.[1,32,36,68,70,71] The exceptions to this are found in sport-specific studies. In gymnastics and figure skating, there is some evidence of an association between poor flexibility and injury.[66,72] Woodford-Rogers[76] also finds that both anterior tibiofemoral laxity and pronation are predictive of anterior cruciate ligament knee injury in adolescents. Pasque and Hewett[39] demonstrate an increased risk of shoulder injury in wrestling with increased shoulder ligament laxity.

There is conflicting evidence that elbow injury in baseball pitchers is related to pitching style.[71,77] Albright et al[77] found an increased risk of elbow injury with a horizontal arm during delivery (particularly with a whipping or snapping motion) in Little League pitchers (≤14 years). Grana and Rashkin[71] found no relationship between injury and sidearm delivery or speed of delivery in older pitchers (14–19 years). Fatigue based on number of pitches in a game and number of pitches in a season seems to be associated with an increased risk of elbow injury.[65] Fatigue also appears to play a role in hockey, where there is an increased risk of injury in the last 5 minutes of a period and the last period of a game.[40] Lysens et al[1] report an increased risk of injury in young women with decreased endurance fitness. This is consistent with the findings of Cahill and Griffith,[29] who report that adolescent football players participating in a preseason conditioning program were at significantly decreased risk of knee injury.

There were only 4 intervention studies addressing prevention of injuries in adolescent sport that were reviewed. These prevention strategies potentially target risk factors such as limitations in flexibility, strength, endurance, and proprioception. Bixler and Jones[68] show no effect of a half-time warm-up and stretching program in high school football in a quasi-experimental, nonrandomized trial. Junge et al[48] demonstrate a significant protective effect of a specific education, conditioning, and rehabilitation program in adolescent soccer players in the low-skilled division only (RR = 0.63 [0.42–0.94]). There were only 2 RCTs examined in this review. Wedderkopp et al[75] demonstrate a significant reduction of injury in adolescent female European handball with the use of a multifaceted training program that included proprioceptive balance training using a wobble board (RR = 0.17 [0.09–0.32]). Heidt et al[64] also demonstrate a protective effect of a multifaceted 7-week preseason training program in female high school soccer players (RR = 0.42 [0.2–0.91]).

Psychosocial factors may also be potentially modifiable. Faelker et al[69] demonstrate evidence of a dose-response gradient between decreasing socioeconomic status and increased risk of injury. Smith et al[73,74] demonstrate a high correlation between injury in sport and stressful life events as well as outcomes on the Short Profile of Mood State (ie, low vigor, high fatigue).

Risk Factors for Injury in Adult Sport

As there are relatively few epidemiological studies addressing modifiable risk factors for injury in child and adolescent sport, it is prudent to discuss epidemiological evidence in adult sport prior to making recommendations for future research. There is inadequate evidence to support decreased muscle strength, globally, as a risk factor for injury in sport. Emery[78] concludes, based on a systematic review of the literature, that there is evidence of an association between decreased hamstring strength and hamstring strain injury in sport. In a review of the literature, Gleim and McHugh[79] find no strong evidence that decreased flexibility is associated with injury in sport. There is significant evidence that decreased sport specific training in the off-season in professional hockey increased the risk of groin strain injury (RR = 3.38 [1.45–7.92]).[80] Poor endurance is a risk factor for injury among army trainees during basic training (RR = 2.8

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TABLE 4. *Studies Examining Risk Factors for Injury in Child and Adolescent Sport*

Study (Year)	Study Design (Country and Time Frame)	Participants (Age, y)	Risk Factor (Exposure Variable)	Injury Definition	Results*
Albright et al[77] (1978)	Cohort (United States, 1973–1974)	109 Little League baseball pitchers (≤14)	(1) Graded pitching form 1 (best form) to 5 (poorest form) based on angle of arm, mechanics, and rhythm (2) Age	Elbow symptoms grades 0 (none) to severe (chronic decreased ROM and marked swelling)	(1) Increased risk of injury with horizontal arm during delivery, particularly with whipping or snapping motion (2) No association between age and injury (insufficient data to calculate RRs)
Backous et al[47] (1988)	Cohort (United States, 5 wk)	1139 Soccer players attending summer soccer camp (6–17)	(1) Sex (female R) (2) Age (3) Height (4) Grip strength	Injury resulting in time loss ≥1 session	(1) RR (male) = 0.69 (0.5 – 0.93) (2) Significantly greater injury rates in ages 14–17, height ≤165 cm, grip strength #25 kg (insufficient data to calculate RRs)
Backx et al[56] (1991)	Cohort (Netherlands, 7 mo, 1982–1983)	1818 School children (8–17) 732 Cases 1032 Controls (all sports)	(1) Venue (outdoor/indoor) (2) Sport activity (high jump rate) (3) Contact (contact/no)	Physical damage caused by sport-related incident	Multiple regression reveals (1) RR (outdoor) = 1.34 (1.19–1.52) (2) RR (high jump rate) = 2.8 (2.42–3.23) (3) RR (contact sport) = 3.03 (2.69–3.42)
Backx et al[7] (1989)	Case-control (Netherlands, 6 wk, 1982)	7468 School children (8–17) 732 Cases 1032 Controls (all sports)	(1) Level of sport activity (low R, moderate, high) (2) Competitive vs. recreational (R) (3) Sex (female R) (4) Age (8–10 R) (5) Elementray (R) vs. secondary (6) Sport (overall R)	Physical damage caused by sport-related incident	(1) OR (mod) = 4.62 (3.16–6.76) OR (high) = 7.2 (4.92–10.54) (2) OR (comp) = 1.75 (3) OR (male) = 1.38 (1.13–1.67) (4) OR (11–12) = 1.33 (1–1.75) OR (13–14) = 2.05 (1.56–2.7) OR (15–16) = 2.09 (1.53–2.83) (5) OR (secondary) = 1.72 (1.41–2.1) (6) OR (basketball) = 1.99 OR (field hockey) = 1.83 OR (track) = 1.54 OR (korfball) = 1.44 OR (handball) = 1.37 OR (soccer) = 1.24 (insufficient data for 95% CIs) 38% reinjury rate
Bijur et al[8] (1995)	Cross-sectional (United States, 1988)	11,840 Children (5–17) (all sports)	(1) Sex all ages (female R) (2) Sex ages 14–17 (female R) (3) Ages (5–9 R)	Medically attended nonfatal injuries occurring in a place of recreation or sports	(1) OR (male) = 1.8 (1.5–2.2) (2) OR (male) = 2.4 (1.8–3.2) (3) OR (10–13) = 1.88 (1.05–3.36) OR (14–17) = 2.23 (1.6–4.78)
Bixler et al[68] (1992)	Quasi-experimental nonrandomized trial (United States)	High school football players (5 teams: 3 intervention, 2 control)	Intervention: half time warm-up and stretching exercises Control: no exercises	Injuries requiring medical attention	Injury rates between groups not statistically significant (insufficient data to calculate RR)
Brust et al[28] (1992)	Cohort (United States, 1991–1992)	150 Boys (9–15) (hockey)	(1) Age (9–13 R) (2) Experience: skill A (R), B, C (3) Weight	Injury in game or practice requiring removal from session	(1) RR (13–15) = 2.45 (1.65–3.63) (2) RR (skill BC) = 2.28 (1.23–4.2) (3) Decreased weight within age group based on student *t* tests ($P < 0.05$) (insufficient data to calculate RR)
Cahill and Griffith[29] (1978)	Historical cohort (United States, 1969–1976)	(1) 1254 High school football players (1969–1972) (2) 2481 High school football players (1973–1976)	(1) Reference group (no preseason conditioning) (2) Exposed group preseason conditioning 80 min/d, 3 times/wk, 4–6 wk)	Knee injury requiring 2 sessions to be missed	Decreased early season injury RR = 0.33 (0.2–0.53) Decreased injury requiring surgery RR = 0.38 (0.16–0.89) Decreased overall injury RR = 0.6 (0.43–0.84)

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TABLE 4—(*Continued*)

Study (Year)	Study Design (Country and Time Frame)	Participants (Age, y)	Risk Factor (Exposure Variable)	Injury Definition	Results*
de Loes[20] (1995)	Cross-sectional (Switzerland, 1987–1989)	689,374 Youth (14–20) (all sports)	(1) Sex (female R) (2) Sex (top 10 sports except soccer R) (3) Sport	Acute sporting injuries	(1) OR (male) = 1.77 (1.07–2.98) (2) OR (male no soccer) = 1.03 (0.61–1.75) (3) Top 3 incidence rates in handball, soccer, basketball
Faelker et al[69] (2000)	Case-control (Canada, 1996)	35,380 Children (<19)	Socioeconomic status (grade 1–5 poverty level, poor/rich) (grade 1 R = highest poverty level)	All injury reporting to emergency	OR (grade 5) = 1.67 (1.48–1.89) OR (grade 5 adolescent only 14–19) = 2.33 (1.92–2.82) Evidence of dose-response with increased injury risk by incresed poverty grade
Garrick and Requa[30] (1978)	Cross-sectional (United States, 1973–1975)	3049 High school sports participants (all sports)	(1) Sex (female R) (2) Sport	Sports injury resulting in removal from session or missing subsequent session	(1) OR (male) = 2.09 (1.83–2.39) (2) Top 3 incidence rates for males in football, wrestling, track; top 3 incidence rates for females in softball, gymnastics, track
Goldberg et al[31] (1988)	Cohort (United States, 1987)	5128 Football players (8–15) (football)	Level of play (youngest and smallest R = junior peewee 8–11/22.5–38.3 kg) (oldest and biggest = bantam 12–15/49.5–67.5 kg)	Injury occurring in football requiring >6 d activity restriction	RR (peewee) = 1.45 (0.79–2.68) RR (junior midget) = 2.92 (1.64–5.19) RR (midget) = 4.45 (2.51–7.88) RR (bantam) = 5.11 (2.53–10.33) 13.1% Reinjury rate
Grace et al[70] (1984)	Cohort (New Mexico)	172 Male high school football players (13–18)	Isokinetic strength imbalance (1) Ipsilateral/ contralateral imbalance ≥10% (2) Hamstring/quads imbalance ≥10% mean	Knee injury requiring missing at least 1 game or practice	No association found between isokinetic strength imbalances 1 or 2 (insufficient data to calculate RR)
Graham and Cleveland[62] (1995)	Cross-sectional (United States, 1994)	634 Junior and senio high school athletes (mean = 13.6)	Left handedness	Injury sustained in sport in the last year requiring physician consult	OR (left) = 2.15 (1.22–3.79)
Grana and Rashkin[71] (1980)	Cohort (United States, 1973–1975)	73 High school baseball pitchers (14–19)	(1) No. seasons played (2) Pitching traits (3) Asymmetry on physical exam (4) Asymmetry on x-ray exam	Elbow symptoms graded 1 (acute episode impairing performance) to 4 (no symptoms or history of injury)	No association between occurrence of symptoms and risk factors 1–4 (insufficient data to calculate RR)
Grubbs et al[32] (1997)	Cohort (United States, 1996)	62 High school basketball players (mean = 15)	Injury score based on Q-angle and weight-bearing asymmetry	Injury requiring missing 1 game or 2 practices	Injury score not predictive of injury Se = 16.7%, Sp = 66.1%, PPV = 5%, NPV = 86.2% No association
Heidt et al[64] (2000)	RCT (United States)	300 Female high school soccer players (14–18)	(1) Intervention (I) 7-week preseason Frappier acceleration program (cardiovascular, plyometrics, strength, and flexibility) (2) Control: no preseason program	Injury requiring missing at least 1 game or practice	RR (I) = 0.42 (0.2–0.91)
Hoff and Martin[33] (1986)	Cohort (United States, 1984–1985)	455 Outdoor and 366 indoor soccer players (U8–U16)	(1) Field type (outdoor R) (2) Age (U10 R)	Injury requiring player to miss next session or limiting playing ability	(1) RR (indoor) = 6.1 (4.17–9.03) (2) RR (10–16) = 2.66 (1.6–4.67)

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TABLE 4—(*Continued*)

Study (Year)	Study Design (Country and Time Frame)	Participants (Age, y)	Risk Factor (Exposure Variable)	Injury Definition	Results*
Junge et al[48] (2002)	Quasi-experimental nonrandomized trial (Switzerland, 1999–2000)	194 Soccer players (mean = 16.5)	(1) Intervention (I) included coach and player education, rehab + conditioning program (F_MARC Bricks including cardiovascular, strength, flexibility, and plyometrics training) (2) Control: ill-defined	Injury resulting in physical complaint >2 wk or missed session	(1) RR (I) = 0.82 (0.58–1.15) (2) RR (I) high skilled divisions = 0.94 (0.58–1.5) (3) RR (I) low-skilled divisions = 0.63 (0.42–0.94)
Kaplan et al[67] (1995)	Cohort (United States)	98 High school football players (mean = 16.6)	(1) Body weight (No. 90 kg R) (2) BMI (No. 75th percentile)	Injury requiring player to miss next session	(1) RR (>90 kg) = 2.53 (1.41–4.55) (2) RR (>75th percentile) = 2.78 (1.05–7.39)
Lenaway et al[21] (1992)	Cohort (United States, 1998–1999)	5518 School students (5–19) (all sports)	Sex (female R)	Injury occurring during sporting activity	RR (male) = 2.13 (1.41–3.21)
Lyman et al[65] (2001)	Cohort (United States, 1997–1998)	298 Baseball pitchers (9–12)	Age, weight, height, no. pitches in game, no. pitches in season, arm fatigue, self-perceived performance, weight lifting, baseball outside league)	Symptoms in the elbow or shoulder during or after a league game	Multivariate analysis (GEE) revealed the following: (1) Risk factors for elbow symptoms: ↑ age, ↑ weight, ↓ height, liftig weights, baseball outside league, ↓ self satisfaction, arm fatigue and throwimg <300 or >600 pitches in season (2) Risk factors for shoulder symptoms: ↓ self satisfaction, arm fatigue, throwing >75 pitches in game, throwing <600 pitches in game, throwing <300 pitches in season
Lysens et al[1] (1984)	Cohort (Netherlands, 1980–1983)	138 Physical education students over 4 y (17–18) (all sports)	Reinjury, endurance fitness, alignment, flexibility, ligamentous laxity	Not well defined	Decreased endurance fitness increased risk of injury in females only ($P < 0.05$) No association between alignment, flexibility, or ligamentous laxity and injury (insufficient data to calculate RRs) 27% Reinjury rate
Maffulli et al[36] (1994)	Cohort (United Kingdom, 1992–1993)	453 Elite athletes (9–18) Football, gymnastics, swimming, tennis	Strength and flexibility	Injury not well defined	No association found between flexibility or strength and injury ($P < 0.05$) (insufficient data to calculate RRs)
McMahon et al[52] (1993)	Cohort (Australia, 1992)	1253 Australian rules football players (U10–U15)	(1) Age (U10 R) (2) Training (R) vs. game (3) Rule modification in U10 only (R) (↓ contact, ↓ field size, ↓ player numbers) vs. conventional (conv)	Injury causing pain or disability; functional impairment (FI) injuries interfered with normal function or consult of a health professional (HP)	(1) RR (U15) = 1.52 (1.17–1.99) RR (FI U15) = 2.2 (1.58–3.1) RR (HP U15) = 7.58 (3.6–18.36) (2) RR (game) = 15.1 (9.3–24.7) (3) RR (conv) = 2.1 (1.2–3.6) for FI injuries only
Messina et al[89] (1999)	Cohort (United States, 1996, 1997)	1863 High school basketball players	Sex (female R)	Injury requiring time loss or physician consult	RR (male) = 1.14 (1.04–1.24) all injury RR (male) = 0.49 (0.36–0.68) knee injury alone
Mueller and Blyth[63] (1974)	Cohort (United States, 1968–1972)	8776 High school football players (13–19)	(1) Previous injury (none R)	Injury requiring time loss ≥1 day or medical attention	(1) RR (previous injury) = 1.43 (1.29–1.58)

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TABLE 4—(*Continued*)

Study (Year)	Study Design (Country and Time Frame)	Participants (Age, y)	Risk Factor (Exposure Variable)	Injury Definition	Results*
Nilsson and Roaas[53] (1978)	Case-control (Norway, 1975–1977)	25,000 soccer players attending international tournament (11–18)	(1) Sex (female R) (2) Age (11–14 R)	All injuries other than minor skin abrasions and blisters reporting to first aid station	(1) RR (male) = 0.52 (0.3–0.88) (2) RR (15–18) = 1.0 (0.56–1.77)
Pasque and Hewett[39] (2000)	Cohort (United States)	418 Male high school wrestlers (14–19)	(1) Ligamentous laxity (2) Age (3) Experience (4) Practice/competition	Injury requiring time loss and trainer or physician consult	(1) ↑ Risk of injury: ↑ age, ↑ experience, competition (2) ↑ Risk of shoulder injury: ↓ ligamentous laxity (insufficient data to calculate RR)
Pinto et al[40] (1999)	Cohort (United States)	22 Junior A hockey players (16–20)	(1) Position (goalie R) vs. forward or defense (2) Practice (R) vs. game (3) Time of season (last ½ R) (4) Time of period and game	Injury requiring medical attention	(1) RR (F or D) = ↓ RR (D game) = 1.1 (2) RR (game) = 75 (3) RR (1st ½ season) = 2.52 (insufficient data to calculate 95% CIs) (4) 46% of injuries in third period, 47% of injuries in final 5 min of period
Powell and Barber-Foss[61] (2000)	Cohort (United States, 1995–1997)	39,032 High school basketball, baseball, and soccer player seasons	Sex (female R)	Injury requiring time loss or medical consult	RR (male) = 0.79 (0.73–0.86) baseball, 0.88 (0.82–0.93) soccer, 0.96 (0.93–1.04) basketball, 0.7 (0.58–0.83) basketball knee injury only
Roberts et al[41] (1999)	Cross-sectional (United States, 1993, 1994)	807 Hockey players (9–19)	Tournament/regular season play (R)	Significant injury resulting in cessation of play, missing next day or medical attention	OR (tournament) = 4–6 (insufficient data to calculate 95% CIs)
Robey et al[42] (1971)	Cohort (United States, 1968)	2252 High school football players (13–19)	(1) Age (13–14 R) (2) Previous season injury (none R)	Injury requiring time loss or medical consult	(1) RR (15) = 1.13 (0.85–1.5) RR (16) = 1.54 (1.17–2.04) RR (≥17) = 2.24 (1.71–2.95) (2) RR (previous injury) = 1.66 (1.35–2.03)
Schmidt-Olsen et al[54] (1985)	Cross-sectional (Denmark, 1984)	6600 Soccer players attending 2 tournaments	Sex (female R)	Injury requiring medical attention	OR male = 0.49 (0.4–0.61)
Smith et al[72] (1991)	Cross-sectional (United States, 1986–1987)	46 Elite junior figure skaters	Flexibility (quadriceps, hamstrings, ITB, ileopsoas)	Anterior knee pain (AKP)	Skaters with AKP had ↓ quadriceps and hamstring flexibility compared to those without AKP ($P < 0.05$) (insufficient data to calculate RR)
Smith et al[73] (1997)	Cohort (United States, 1996)	86 Male high school ice hockey players	Physical, situational, and psychosocial factors	Injury requiring medical attention or time loss >24 h	High playing time, low vigor, and high fatigue as per Short Profile of Mood States significantly predicted injury (insufficient data to calculate RRs)
Smith et al[74] (1990)	Cohort (United States, 1988)	451 High school athletes	Psychosocial factors (stressful life events, social support, coping skills)	Injury requiring time loss >1 d	Stressful life events accounted for 30% of injury variance in subjects with low social support and coping skills (insufficient data to calculate RRs)
Sorensen et al[25] (1996)	Cross-sectional (Denmark, 1988–1992)	63,017 School-age children (6–17) (all sports)	(1) Sex (female R)	Injury resulting from sport activity presenting to the hospital emergency department	(1) OR (male) = 1.13 (1.07–1.18)

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TABLE 4—(*Continued*)

Study (Year)	Study Design (Country and Time Frame)	Participants (Age, y)	Risk Factor (Exposure Variable)	Injury Definition	Results*
Steele and White[43] (1986)	Cohort (Britain, 1985)	40 Elite female gymnasts (10–21)	Anthropometric variables, hypermobility, flexibility, and spinal curvature	Injury score 1 (head injury, fracture, or dislocation) to 8 (minor injury: modified training <1 week)	Multiple regression yields accurate prediction of high (40+) and low (0–15) injury score with 5 variables: (1) lordosis (2) mesomorphy (3) weight (4) age (5) height
Stuart et al[44] (2002)	Cohort (United States, 1997)	915 Football players (9–13)	(1) Age (school grade 4 R) (2) Weight	Injury preventing participation and requiring medical attention	(1) RR (grade 5) = 2.07 (1.43–2.71) RR (grade 6) = 2.31 (1.59–2.74) RR (grade 7) = 2.19 (1.56–2.83) RR (grade 8) = 4.13 (3.49–4.77) (2) Weight not predictive of injury
Sullivan et al[45] (1980)	Cohort (United States)	1272 Soccer players (7–18)	(1) Sex (female R) (2) Age (U14 R)	Injury preventing participation	(1) RR (male) = 0.46 (0.22–0.98) (2) RR (14–18) = 9.3 (4.29–21.78)
Wedderkopp et al[75] (1999)	RCT (Denmark, 1995, 1996)	237 Female European team handball players (16–18)	(1) Intervention: practice session training program (warm-up with 2 or more functional large muscle group exercises and proprioceptive ankle disk activity) (2) Control: nonspecific practice session training	Injury requiring player to miss next session or unable to participate without considerable discomfort	RR (intervention) = 0.17 (0.09–0.32)
Woodford-Rogers et al[76] (1994)	Matched case-control (United States)	44 Football players, basketball players, and gymnasts	Pronation and anterior knee joint laxity measurement	ACL knee ligament injury (complete tear)	Discriminant analysis correctly predicts 87.5% all females and 70.5% all cases (insufficient data to calculate ORs)
Wright and De Cree[66] (1998)	Cross-sectional (United Kingdom)	15 Female competitive gymnasts (8–18)	(1) Age (2) Height (3) Mass (4) Somatotype (5) Years of experience (6) Strength (7) Flexibility	4-Year injury history (classified as high or low injury status based on no. injuries and time loss	High injury group significant ↑ age, ↑ height, ↑ mass, ↑ years experience and ↑ flexibility (ankle dorsiflexion and back extension) in univariate analysis only Strength and somatotype not significant
Yde and Nielsen[55] (1990)	Cohort (Denmark, 1985–1986)	302 Soccer, handball, and basketball players (<18 y)	(1) Age (<14 R) (2) Sport played (handball + basketball R)	Injury requiring player to miss next session	(1) RR (≥14) = 3.14 (2.49–4.77) (2) RR (soccer <14) = 6.47 (2.06–20.33) RR (soccer ≥14) = 1.79 (1.35–2.37)
Zaricznyj et al[24] (1980)	Cross-sectional (United States, 1974–1975)	25,512 School-age children in community (all sports)	(1) Sex (female R) (2) Junior high (R) vs. high school (3) Organized (R) vs. physical education and nonorganized sports	Injury occurring in sport requiring medical treatment or filing of school/insurance forms	(1) OR (male) = 2.1 (1.9–2.33) (2) OR (high school) = 2.74 (2.48–3.02) (3) OR (PE + nonorganized) = 0.4 (0.35–0.46)

*RR and OR provided if adequate information provided. OR indicates odds ratio; R, reference exposure group; RCT, randomized clinical trial; RR, relative risk.

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[1.2–6.7] for men and 1.69 [1.2, 2.4] for women).[34] Previous injury appears to be the most significant predictor of sports injury in some studies, with relative risks ranging from 2.88 to 9.41.[80–82] Tropp et al[83] demonstrate that soccer players with functional ankle instability and decreased balance ability were at significantly greater risk of ankle sprain reinjury.

Some studies have examined proprioceptive balance training in conjunction with other training strategies (ie, strengthening, endurance training, plyometrics) to reduce injury in sport. These multifaceted training programs have been shown to reduce significantly the incidence of ankle sprain injuries and anterior cruciate ligament injuries in some sports.[26,75,84,85] However, balance, endurance, and strength have not been examined as outcome measurements, so the impact of the training strategies on these potential risk factors is not clear.

DISCUSSION

Nonmodifiable Risk Factors

Male children and adolescents participating in sport may be at a greater risk of injury as they may be more aggressive, have larger body mass, and experience greater contact compared with girls in the same sports. All of these factors may lead to increased forces in running, jumping, pivoting, and contact, which may increase susceptibility to injury. In soccer, baseball, and basketball, studies show an increased risk of injury in girls. The reason for this may be lower skill level, or it may be physiological. It is hypothesized that left-handed adolescents may be at increased risk of injury because of environmental biases in a right-handed world (ie, equipment used in sport) or functional differences related to neurologic development.[62] Previous injury clearly increases the risk of injury in sport. This finding may be related to persistent symptoms, underlying physiological deficiencies resulting from the initial injury (ie, ligamentous laxity, muscle strength, endurance, proprioception), or inadequate rehabilitation.

It is not surprising that hockey, basketball, and football are consistently among the top-rated sports for injury in male athletes. There is certainly significant contact involved in 2 of the 3 sports (hockey and football) and some contact in basketball also. All 3 sports involve a high rate of jumping, sprinting, and pivoting activity often involved in the mechanism of injury in sport. The findings of Backx et al[56] of outdoor sports, high jump rate sports, and contact sports increasing the risk of injury are consistent with the high rates of injury in these 3 sports. It is also not surprising that gymnastics, basketball, and soccer are consistently among the top-rated sports for injury in female athletes. These 3 sports also involve a high rate of jumping, sprinting, and pivoting activity.

Consistency in the findings that risk of injury in child and adolescent sport increases with age is not surprising given that level of competition, contact, and size typically increase with age. Time participating in sport also likely increases with age and experience; however, exposure-adjusted injury rate (incidence density) is often not examined. Taller and heavier athletes (ie, in football, gymnastics, soccer, and baseball) may be more susceptible injury due to greater forces being absorbed through soft tissue and joints. In hockey, a contact sport in which there is no weight classification, it is not surprising that the smaller players are more susceptible to injury.

Injury reporting may be more accurate in studies examining organized sport (ie, levels of competition) and tournament play, accounting for higher injury rates than in unorganized sport. In addition, competitors are more likely to be playing at greater intensity and speeds in competition and tournaments than in practice and regular season play, increasing the risk of sustaining an injury. In Australian rules football, it is not surprising that larger field size and fewer players (ie, likely reducing the risk of contact) appear to be associated with a lower risk of injury.[52] With rapid skeletal growth occurring in children and adolescents, there are potentially physiological reasons why children and adolescents may be at increased risk of injury.[60] For example, sudden intense muscular traction exerted on an immature skeleton (ie, during a period of rapidly increasing muscular strength) may result in an acute avulsion fracture of a growth plate, an injury not possible in adulthood.[60] Chronic repetitive muscular traction exerted on an immature skeleton, usually at the time of a growth spurt, may result in traction apophysitis (ie, Osgood-Schlatter or Sever disease).[60] These are both injuries exclusive to children and adolescents.

Potentially Modifiable Risk Factors

Upton et al[86] demonstrate that less than 40% of high school Rugby participants (n = 2330) completed any preseason training. High rates of injury may be related to decreased endurance or strength associated with limited preseason training, as indicated in both adolescent[1,29,48,64,75] and adult[34,80,84] study findings. Some athlete populations (ie, low-skill division adolescent female soccer players) may benefit from training programs, while others (ie, high-skill division adolescent female soccer players) may not.[48] Decreased flexibility does not appear to be a risk factor generally for injury in adolescent[1,32,36] or adult sport.[79] However, specifically, it may be a risk factor for injury in gymnastics and figure skating, both sports that demand a high degree of flexibility for execution of many maneuvers.[66,72] Proprioceptive balance training, in conjunction with other training techniques, may reduce the risk of specific injury in specific sports.[75,84,85] The impact of decreased proprioception as a risk factor for injury remains unclear. The findings that psychosocial factors (ie, low socioeconomic status and high stressful life events) increase the risk of child and adolescent injury in sport are also consistent with the findings for other injury types (ie, home, fall, and traffic injury).[69,73,74]

Study Limitations

One of the fundamental difficulties in comparing research in sports injury epidemiology is the variability in research design and measurements used to assess expo-

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sure and injury and the variety of risk factors and sports assessed in studies. The research designs reviewed are almost exclusively observational, and intervention studies are not always RCTs. The temporal association between exposure and outcome is often ignored in cross-sectional and case-control studies. For example, Smith et al[72] examine flexibility in figure skaters already presenting with knee pain, and the temporal association between knee pain and decreased flexibility is unclear.

Injury definition and methods of injury data collection are extremely variable. One of the major limitations in many studies reviewed is that incidence rates based on number of participants rather than incidence densities based on exposure (ie, hours or sessions of participation) are used to distinguish high-risk athletes. Clearly, time spent doing an activity is critical in the assessment of risk of injury. Time loss, medical requirements, and reinjury inclusion differ widely among injury definitions. Methods of data collection vary from self-report to therapist or physician report. Only 25% to 31% of injuries in some studies resulted in a physician consult.[7,37,56] Depending on injury definition, some studies may underestimate injury if only those reporting to an emergency room,[13,15,18,25] physician, or therapist[40,54] are included. Other studies may overestimate injury rates if all injuries are reported, regardless of reporting source (ie, parent, coach).[7,56] If one relies on self-report, particularly over a longer period, incidence rates will likely be underestimated due to recall bias. Bijur et al[8] demonstrate a 51% increase in self-reported injury over a 1-month recall period compared with a 12-month recall period.

Selection bias is of concern in many studies as there is no random selection of participants. Selection bias in which athletes selected are more likely to be injured (ie, previous injury) and more likely to be in the exposure risk group may lead to an overestimation of association between risk factor and injury.[1,29,43,66,69,72–74,76] If there are unreported dropouts from the study and the reason for dropout is related to injury, this may lead to an underestimation of association (another form of selection bias).[1,32,36,68,71] Lack of blinding to exposure status, as with most of the cohort studies examined in this review, may also lead to overestimation of the association.

Poor reliability and validity of exposure measurements (ie, flexibility, strength) resulting in nondifferential misclassification of exposure (ie, likelihood of misclassification of exposure is not associated with outcome) will underestimate the association between exposure and injury. This is certainly of concern in studies that found no association.[1,32,36,68,71]

The most significant source of bias in the studies reviewed was a lack of measurement and control for potentially confounding variables. This results most often in an overestimation of association between exposure and injury. When recruitment of subjects is not random, risk factors or training interventions assessed may not be the only difference between groups. Differences in physiological factors, coaching technique, warm-up routines, and equipment may prevail. For example, in the study by Cahill,[29] a historical cohort, differences attributed to preseason conditioning may be a result of equipment differences, coaching differences, rule changes (ie, elimination of below-the-waist blocking in 1973),[87] or physiologic factors in the 2 cohorts that were not controlled for in the study.

In some RCT studies examining prevention strategies, the intervention was assigned to a team (ie, cluster), not an individual.[48,64,75] Similarities within a team are likely greater than similarities between teams; these similarities should be controlled for in the analysis (ie, cluster analysis). When clusters are controlled for in an analysis, the effect measure is less precise (ie, larger 95% CIs) if similarities within each cluster are in fact greater than similarities between clusters.[88] As such, overestimates of the protective effects of training strategies may have been reported as a result of the individual level analyses done in these intervention studies. In addition, the intervention studies examined identify multifaceted preventative training programs.[48,64,68,75] As a result, it is difficult to identify specific risk factors addressed by the program (ie, flexibility, strength, endurance, proprioception) if measurements of these factors are not examined.

External validity of the results in all of the studies examined is limited due to limitations in internal validity. Certainly, generalizability beyond the specific sport, age group, level of competition, and specific injury type is limited.

In examining the Hill criteria of causation,[59] many of the studies reviewed are consistent with the findings in adult population studies. The strength of the associations found between preparticipation training programs and injury is convincing based on the magnitude of associations found, despite concerns with internal validity and individual-level analysis. Specificity, implying that a specific cause leads to a specific effect, is difficult to identify when studies often do not control for other risk factors and injury outcome is often global and poorly defined. Temporal association is clear only in the cohort studies and RCTs reviewed. The only studies providing a clear indication of a dose-response relationship is that by Faelker et al,[69] in which injury rate increases with increasing level of poverty, and the studies examining increased risk of injury with increasing age.[7,8,31,42,43,65] Biologic plausibility of risk factors and coherence to existing knowledge have been discussed. Injury prevention studies are few; thus, experimental evidence is certainly limited.

CONCLUSIONS

Child and adolescent participation rates in sport are high. High rates of sports injury in this population result in a significant impact on the individual, the parents, and the health care system. Sports injury in children and adolescents may also potentially affect future involvement in physical activity and the future health of our population.

The strength of the evidence for potentially modifiable risk factors for injury in children and adolescents is limited by research design and concerns with internal validity. In case-control and cross-sectional study designs, the

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temporal association between exposure and outcome is unclear. In many of the cohort studies and nonrandomized intervention studies reviewed, various sources of bias in the selection of subjects, measurement of exposure and outcome variables, and lack of control for other potentially confounding variables threaten the studies' internal validity. There is limited RCT evidence supporting preventative training programs in specific sports in adolescents to reduce the risk of injury. There is certainly more convincing evidence in adult epidemiological studies that decreased endurance, decreased strength, decreased proprioception, and decreased preseason sport-specific training are associated with sports injury. The consistency of the findings between child and adolescent studies reviewed and the adult population studies is encouraging.

The results of this review can be used in targeting relevant athlete groups (ie, high-risk sports such as hockey, basketball, football, soccer [particularly indoor], and gymnastics), age groups (ie, older adolescents), and skill levels (ie, low-skill division in female adolescent soccer) in designing future research examining risk factors and prevention strategies in child and adolescent sport. Future studies examining prevention strategies such as preseason conditioning and proprioceptive training are warranted. Future clinical trials examining such prevention strategies should quantify and control for potential risk factors for injury in child and adolescent sport.

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